Case Study II: Styrian Health Data

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What we did last week

- Cleaning and preprocessing of the data
- Exploratory data analysis

Overview and Cleaning

- Dropped `postalcode`, `municipality` and `district` columns
- Column-wise null values are:
  - `bundesland` – 331
  - `befinden` – 23
  - `geburtsjahr` – 23
  - `geschlecht` – 23
  - `schaetzwert_bp_sys` – 45
  - `schaetzwert_by_dia` – 56
- Replaced estimated values of blood pressure with mean values
- Added a new column age

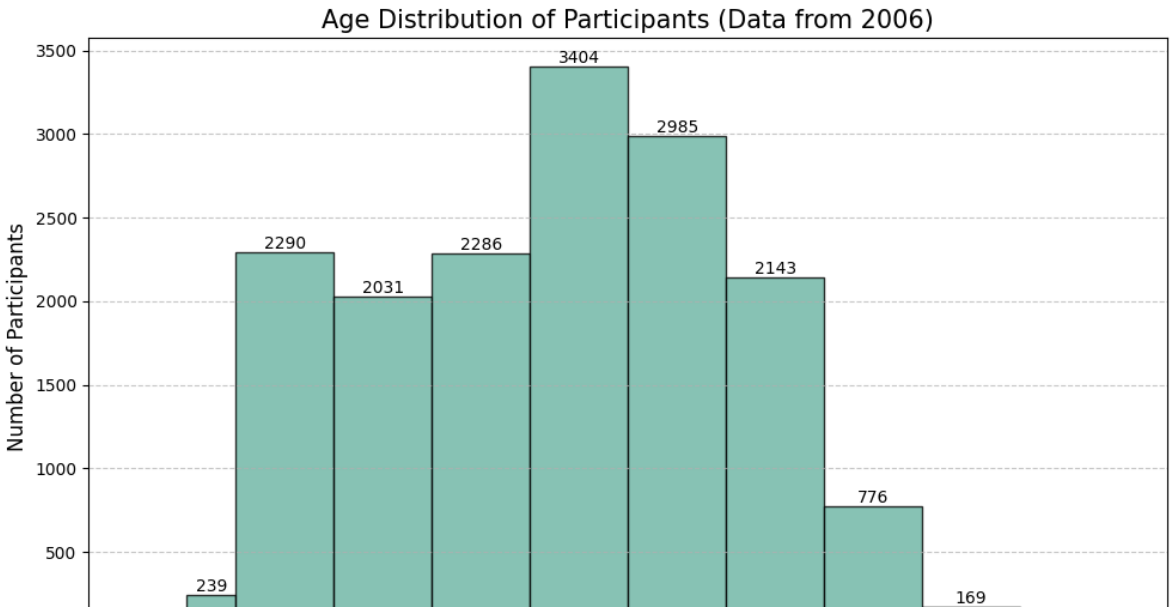
Analysis 1.1

- Interesting pattern found in all the records where `befinden` is null!

	id	zeit	terminal	bundesland	befinden	geburtsjahr	geschlecht	raucher	blutzucker_bekannt	cholesterin_bekannt	in_behandlung
2594	14993	2006-05-30 14:36:39	3	NaN	NaN	NaN	NaN	False	False	False	False
2599	15011	2006-05-30 15:21:46	3	NaN	NaN	NaN	NaN	False	False	False	False
2656	15158	2006-05-31 16:57:53	3	NaN	NaN	NaN	NaN	False	False	False	False
2660	15172	2006-05-31 19:42:51	3	NaN	NaN	NaN	NaN	False	False	False	False
2690	15247	2006-06-01 13:29:30	3	Niederösterreich	NaN	NaN	NaN	False	False	False	False

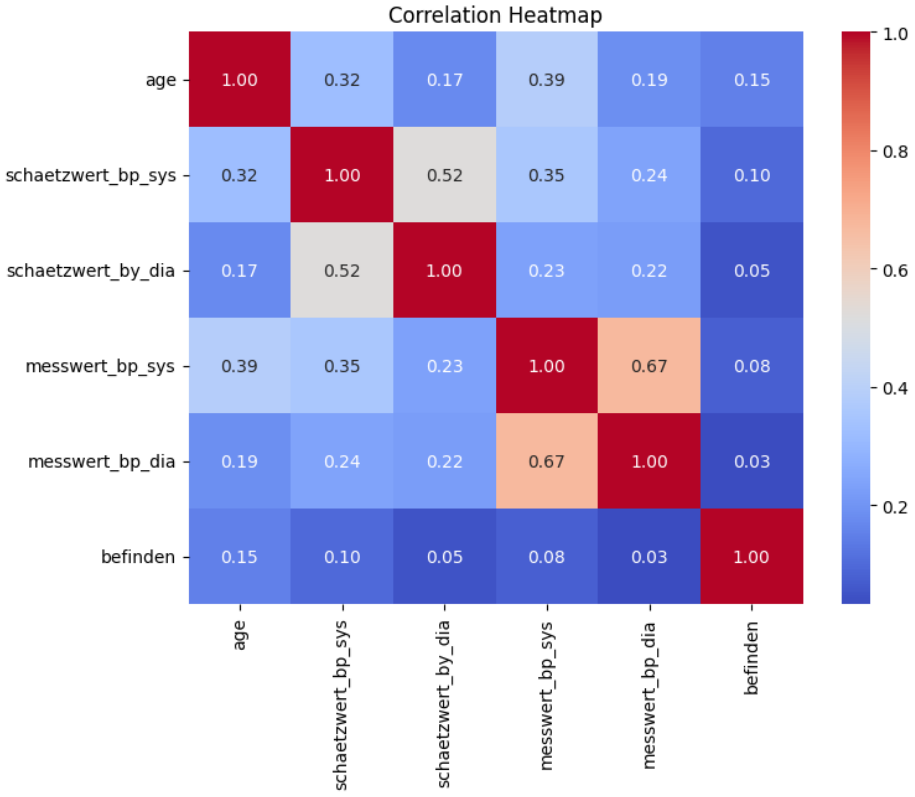
Analysis 1.2

- Age distribution
- 73 smokers out of 1700 children below 15 years of age!
- Suprisingly estimates and realities are close for people over 100 years of age



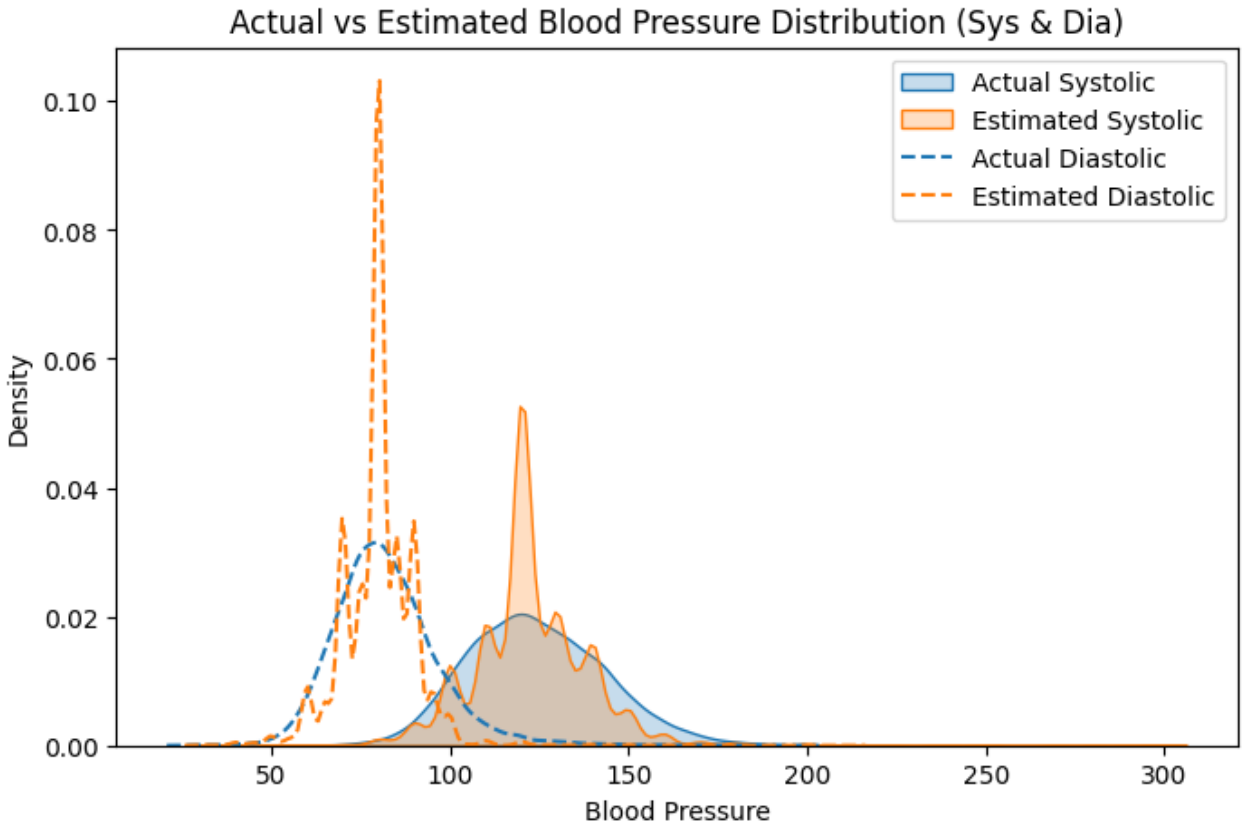
Analysis 1.3

- Correlation between columns
- Measured BP values have significant correlation.



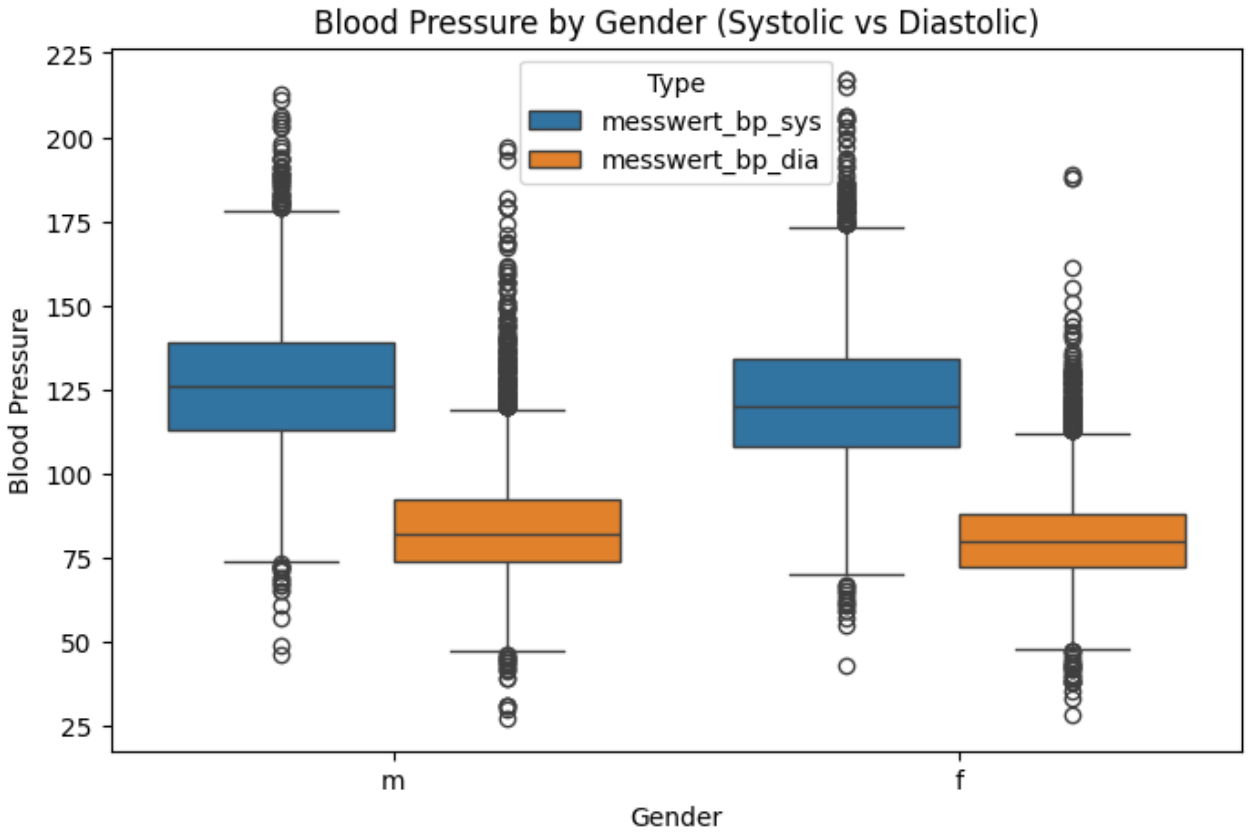
Analysis 1.4

- Difference between assumption and reality



Analysis 1.5

- Gender comparison



Agenda for the this week

- Try combinations of columns and rows and see whether overall distribution changes
- Discussing whether its a classification or regression problem
- Fitting different models and evaluating them